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## **CMPG325-COMPUTER NETWORKS**

### **Milestone 2: CLIENT IMPLEMENTATION REVIEW**

|                              |                                              |
|------------------------------|----------------------------------------------|
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| <b>Project ID</b>            | CMPG325-2026-099                             |
| <b>Client ID</b>             | CLI-099                                      |
| <b>Assigned Organization</b> | Mafikeng Local Municipality Satellite Office |
| <b>Industry</b>              | Municipal Services                           |
| <b>Repository</b>            | <a href="#"><u>GITHUB</u></a>                |

## 1. Introduction

Milestone 1 presented the proposed network design for the Mafikeng Local Municipality Satellite Office, including the addressing scheme, VLAN structure, and the routing and security approach selected to meet the client's requirements. This document, Milestone 2, records the implementation of that design in Cisco Packet Tracer and presents the testing evidence collected to verify that the implemented network functions as intended.

The report does not repeat the Milestone 1 design discussion. It focuses on confirming that the network has been built, that the assigned technical challenge (static routing with multi-router path control) operates correctly, that the client change request (CR8 — shared printer zone) has been implemented, and that the client's security constraint restricting Public Counter access to the Administration network is enforced.

## 2. Implemented Network Overview

The implemented network consists of three routers connected in series — MUN-R1, MUN-R2, and MUN-R3 — supporting five VLANs across three access-layer switches. The assigned IPv4 block 192.168.43.0/24 is subnetted to provide each VLAN and inter-router link with its own address space.

### 2.1 Topology

- MUN-R1 serves VLAN 10 (Public Counter) and VLAN 20 (Licensing & Permits).
- MUN-R2 serves VLAN 50 (Shared Printer Zone) and acts as the intermediate router between MUN-R1 and MUN-R3.
- MUN-R3 serves VLAN 30 (Municipal Administration) and VLAN 40 (Finance).
- MUN-SW1, MUN-SW2, and MUN-SW3 provide access-layer switching and 802.1Q trunking to their respective routers.

### 2.2 Core Implementation Elements

- Static routing is configured on all three routers to provide the required inter-network reachability across the implemented subnets, in line with the assigned Multi-Router Path Control challenge and subject to the configured security policy.
- CR8 — a shared printer zone (VLAN 50) — is centralized behind MUN-R2 and made reachable from VLAN 20 (Licensing & Permits) and VLAN 30 (Municipal Administration).
- A named extended ACL, PUBLIC\_COUNTER\_RESTRICTION, is applied on MUN-R1 to enforce the client's requirement that Public Counter staff must not access the Administration network.

### 3. VLAN and Trunk Implementation

Each access switch was verified using show vlan brief and show interfaces trunk to confirm VLAN membership, port assignment, and 802.1Q trunk operation toward the corresponding router.

#### 3.1 MUN-SW1

MUN-SW1 hosts VLAN 10 (CITIZEN\_SERVICES), used for the Public Counter department, on ports Fa0/1–Fa0/3, and VLAN 20 (LICENSING\_PERMITS) on ports Fa0/4–Fa0/6. Port Fa0/24 is configured as an 802.1Q trunk in the "on" mode, actively forwarding VLANs 10 and 20 to MUN-R1.



```

MUN-SW1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

MUN-SW1>enable
MUN-SW1#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Gig0/1, Gig0/2
10   CITIZEN_SERVICES        active    Fa0/1, Fa0/2, Fa0/3
20   LICENSING_PERMITS        active    Fa0/4, Fa0/5, Fa0/6
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

MUN-SW1#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    10,20

Port      Vlans allowed and active in management domain
Fa0/24    10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20

MUN-SW1#

```

Figure 1: MUN-SW1 VLAN Configuration and Fa0/24 Trunk to MUN-R1

### 3.2 MUN-SW2

MUN-SW2 hosts VLAN 50 (SHARED\_PRINTER\_ZONE), with the shared printer PRINT-01 connected through Fa0/1. Port Fa0/2 is configured as an 802.1Q trunk carrying VLAN 50 to MUN-R2, confirming that the printer zone is correctly isolated and trunked to its gateway router.

```

MUN-SW2
Physical Config CLI Attributes
Switch Ports Model          SW Version  SW Image
-----
*    1 26    WS-C2960-24TT-L   15.0(2)SE4   C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

MUN-SW2>enable
MUN-SW2#show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2
50   SHARED_PRINTER_ZONE    active    Fa0/1
1002 fddi-default          active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

MUN-SW2#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/2     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/2     50

Port      Vlans allowed and active in management domain
Fa0/2     50

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/2     50

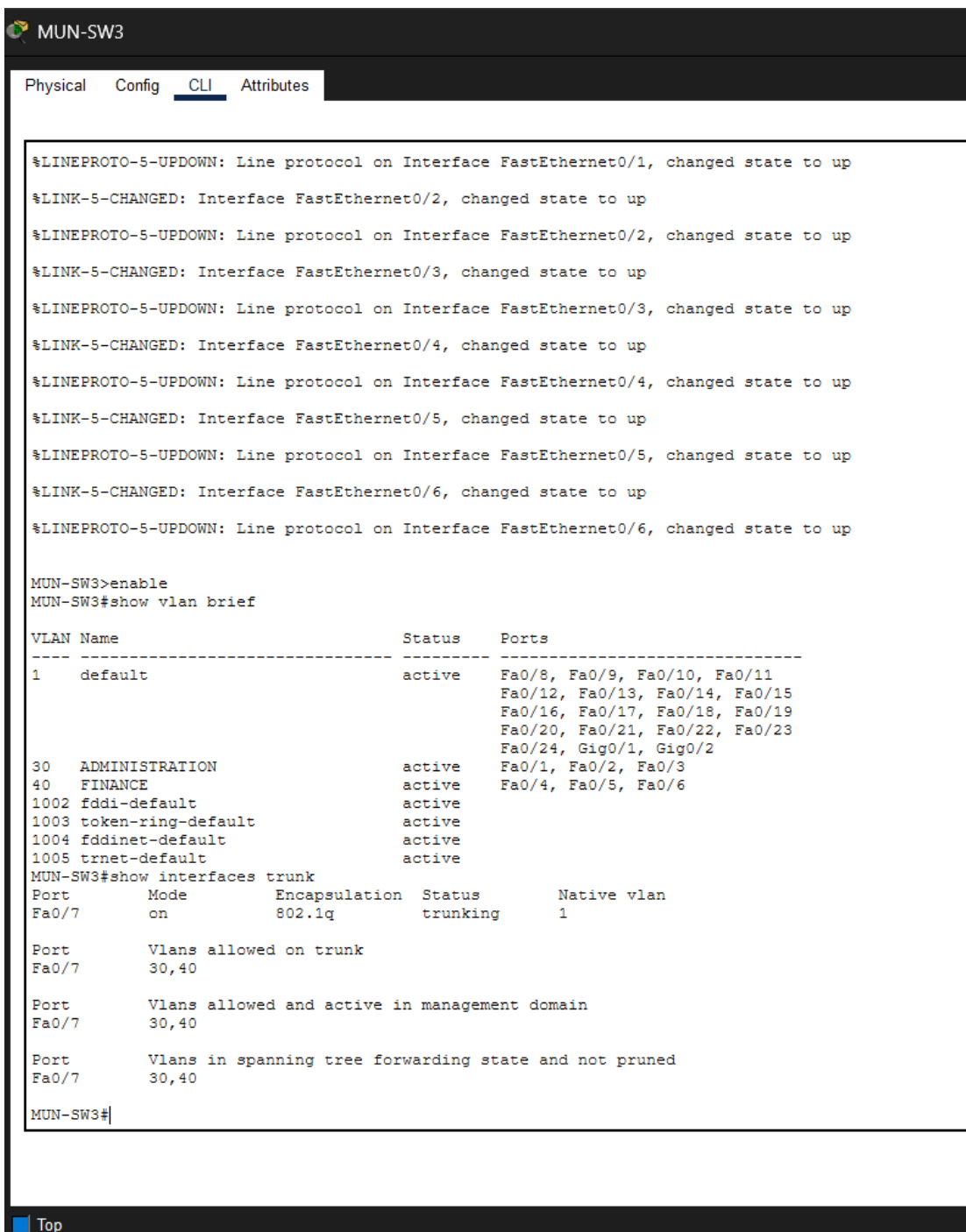
MUN-SW2#

```

Figure 2: MUN-SW2 Shared Printer VLAN and Fa0/2 Trunk to MUN-R2

### 3.3 MUN-SW3

MUN-SW3 hosts VLAN 30 (ADMINISTRATION) on ports Fa0/1–Fa0/3 and VLAN 40 (FINANCE) on ports Fa0/4–Fa0/6. Port Fa0/7 is configured as an 802.1Q trunk carrying VLANs 30 and 40 to MUN-R3.



```

MUN-SW3
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

MUN-SW3>enable
MUN-SW3#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
30   ADMINISTRATION         active    Fa0/1, Fa0/2, Fa0/3
40   FINANCE                 active    Fa0/4, Fa0/5, Fa0/6
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

MUN-SW3#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/7     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/7     30,40

Port      Vlans allowed and active in management domain
Fa0/7     30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/7     30,40

MUN-SW3#
  
```

Top

Figure 3: MUN-SW3 VLAN Configuration and Fa0/7 Trunk to MUN-R3

## 4. Static Routing Implementation

The routing table on each router was inspected using `show ip route`, with attention restricted to the S (static) entries, since these are the routes relevant to the assigned technical challenge.

### 4.1 MUN-R1

MUN-R1's routing table shows static routes to 192.168.43.64/27, 192.168.43.96/27, and 192.168.43.128/28, each reached via next hop 192.168.43.146 (MUN-R2). This confirms MUN-R1 correctly forwards traffic destined for the Administration, Finance, and Shared Printer Zone networks toward MUN-R2.

```

MUN-R1
Physical Config CLI Attributes

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

MUN-R1>enable
MUN-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.43.0/24 is variably subnetted, 10 subnets, 4 masks
C       192.168.43.0/27 is directly connected, GigabitEthernet0/0.10
L       192.168.43.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.43.32/27 is directly connected, GigabitEthernet0/0.10
C       192.168.43.32/27 is directly connected, GigabitEthernet0/0.20
L       192.168.43.33/32 is directly connected, GigabitEthernet0/0.20
S       192.168.43.64/27 [1/0] via 192.168.43.146
S       192.168.43.96/27 [1/0] via 192.168.43.146
S       192.168.43.128/28 [1/0] via 192.168.43.146
C       192.168.43.144/30 is directly connected, Serial0/0/0
L       192.168.43.145/32 is directly connected, Serial0/0/0
S       192.168.43.148/30 [1/0] via 192.168.43.146

MUN-R1#
  
```

Figure 4: MUN-R1 Routing Table Showing Static Routes via MUN-R2

## 4.2 MUN-R2

MUN-R2's routing table shows static routes to 192.168.43.0/27 and 192.168.43.32/27 via 192.168.43.145 (MUN-R1), and to 192.168.43.64/27 and 192.168.43.96/27 via 192.168.43.150 (MUN-R3). This confirms MUN-R2's role as the intermediate router, forwarding traffic in both directions between MUN-R1 and MUN-R3.



```

MUN-R2
Physical Config CLI Attributes

http://www.cisco.com/wai/export/crypto/colo/stqlg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K5
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

MUN-R2>enable
MUN-R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

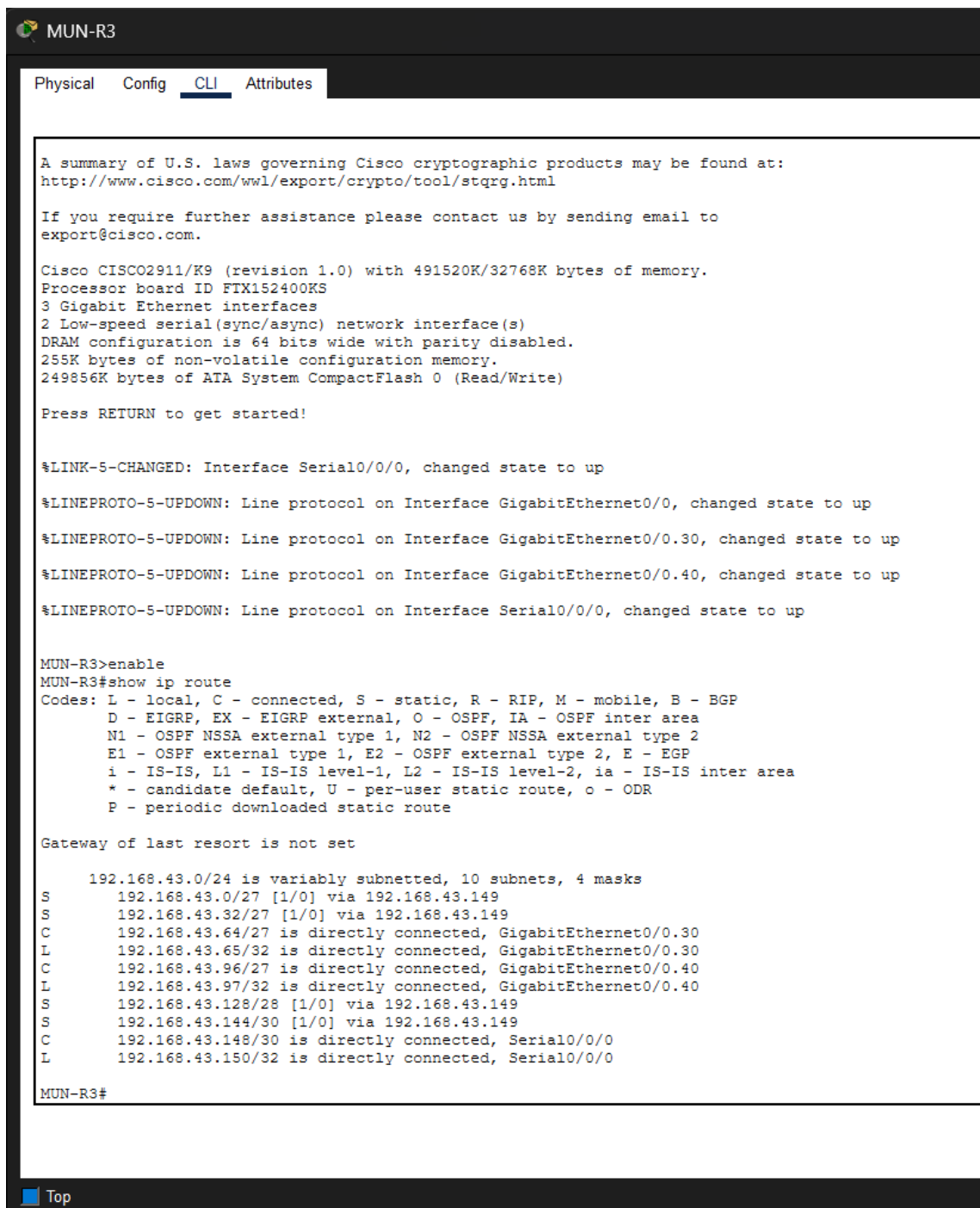
    192.168.43.0/24 is variably subnetted, 10 subnets, 4 masks
S       192.168.43.0/27 [1/0] via 192.168.43.145
S       192.168.43.32/27 [1/0] via 192.168.43.145
S       192.168.43.64/27 [1/0] via 192.168.43.150
S       192.168.43.96/27 [1/0] via 192.168.43.150
C       192.168.43.128/28 is directly connected, GigabitEthernet0/0.50
L       192.168.43.129/32 is directly connected, GigabitEthernet0/0.50
C       192.168.43.144/30 is directly connected, Serial0/0/0
L       192.168.43.146/32 is directly connected, Serial0/0/0
C       192.168.43.148/30 is directly connected, Serial0/0/1
L       192.168.43.149/32 is directly connected, Serial0/0/1

MUN-R2#
  
```

Figure 5: MUN-R2 Routing Table Confirming Its Role as Intermediate Router

### 4.3 MUN-R3

MUN-R3's routing table shows static routes to 192.168.43.0/27 and 192.168.43.32/27, each via next hop 192.168.43.149 (MUN-R2). This confirms MUN-R3 correctly forwards traffic destined for the Public Counter and Licensing & Permits networks back toward MUN-R2.



```

MUN-R3
Physical Config CLI Attributes

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

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3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

MUN-R3>enable
MUN-R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.43.0/24 is variably subnetted, 10 subnets, 4 masks
S       192.168.43.0/27 [1/0] via 192.168.43.149
S       192.168.43.32/27 [1/0] via 192.168.43.149
C       192.168.43.64/27 is directly connected, GigabitEthernet0/0.30
L       192.168.43.65/32 is directly connected, GigabitEthernet0/0.30
C       192.168.43.96/27 is directly connected, GigabitEthernet0/0.40
L       192.168.43.97/32 is directly connected, GigabitEthernet0/0.40
S       192.168.43.128/28 [1/0] via 192.168.43.149
S       192.168.43.144/30 [1/0] via 192.168.43.149
C       192.168.43.148/30 is directly connected, Serial0/0/0
L       192.168.43.150/32 is directly connected, Serial0/0/0

MUN-R3#
  
```

Figure 6: MUN-R3 Routing Table Showing Static Routes via MUN-R2

## 5. Assigned Technical Challenge — Multi-Router Path Control

The assigned technical challenge, Static Routing — Multi-Router Path Control, was verified using a traceroute from LP-PC01 (VLAN 20, Licensing & Permits) to ADM-PC01 (VLAN 30, Municipal Administration) at 192.168.43.66.

```

LP-PC01
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.43.66

Tracing route to 192.168.43.66 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.43.33
  2  1 ms    0 ms    0 ms    192.168.43.146
  3  1 ms    0 ms    0 ms    192.168.43.150
  4  *        1 ms    1 ms    192.168.43.66

Trace complete.

C:\>

```

Figure 7: Multi-Router Path Verification from Licensing & Permits to Municipal Administration

Figure 7 shows a traceroute from LP-PC01 to 192.168.43.66. The first hop, 192.168.43.33, is the VLAN 20 default gateway on MUN-R1. The second hop, 192.168.43.146, is MUN-R2's interface facing MUN-R1. The third hop, 192.168.43.150, is MUN-R3's interface facing MUN-R2. The fourth hop reaches the destination, ADM-PC01, at 192.168.43.66.

This hop sequence is direct evidence of the assigned challenge rather than a simple confirmation of reachability: a traceroute reveals the actual Layer 3 path taken by each packet, hop by hop, whereas a ping only confirms that a destination is reachable without disclosing the path. The observed sequence — MUN-R1 → MUN-R2 → MUN-R3 — demonstrates that traffic is being forwarded across all three routers strictly according to the configured static routes, with MUN-R2 correctly performing its role as the intermediate hop between the two edge routers. This confirms that multi-router path control has been correctly implemented using static routing across the full three-router topology.

## 6. Client Change Request CR8 — Shared Printer Zone

CR8 required that a shared printer zone be made available to two departments that previously could not print: Licensing & Permits (VLAN 20) and Municipal Administration (VLAN 30). The shared printer, PRINT-01 (192.168.43.130/28, gateway 192.168.43.129), is centralized behind MUN-R2 in VLAN 50, so that both departments reach it via the routed path through their respective routers rather than requiring a local printer in each department.

### 6.1 Licensing & Permits Access

LP-PC01 was tested against PRINT-01. Figure 8 shows the traceroute confirming the routed path to 192.168.43.66, followed by two ping tests to 192.168.43.130. The first ping recorded one timeout (Sent = 4, Received = 3, 25% loss), which is attributable to initial ARP address resolution rather than a routing or configuration fault. The subsequent ping to the same destination returned a clean result of Sent = 4, Received = 4, Lost = 0 (0% loss), confirming stable IP reachability from Licensing & Permits to the shared printer.

```

LP-PC01
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.43.66

Tracing route to 192.168.43.66 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.43.33
  1  1 ms    0 ms    0 ms    192.168.43.146
  2  1 ms    0 ms    0 ms    192.168.43.150
  3  *        1 ms    1 ms    192.168.43.66

Trace complete.

C:\>ping 192.168.43.130

Pinging 192.168.43.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.43.130: bytes=32 time=2ms TTL=126
Reply from 192.168.43.130: bytes=32 time=1ms TTL=126
Reply from 192.168.43.130: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.43.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>cls
Invalid Command.

C:\>ping 192.168.43.130

Pinging 192.168.43.130 with 32 bytes of data:

Reply from 192.168.43.130: bytes=32 time=1ms TTL=126
Reply from 192.168.43.130: bytes=32 time=24ms TTL=126
Reply from 192.168.43.130: bytes=32 time=2ms TTL=126
Reply from 192.168.43.130: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.43.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 7ms

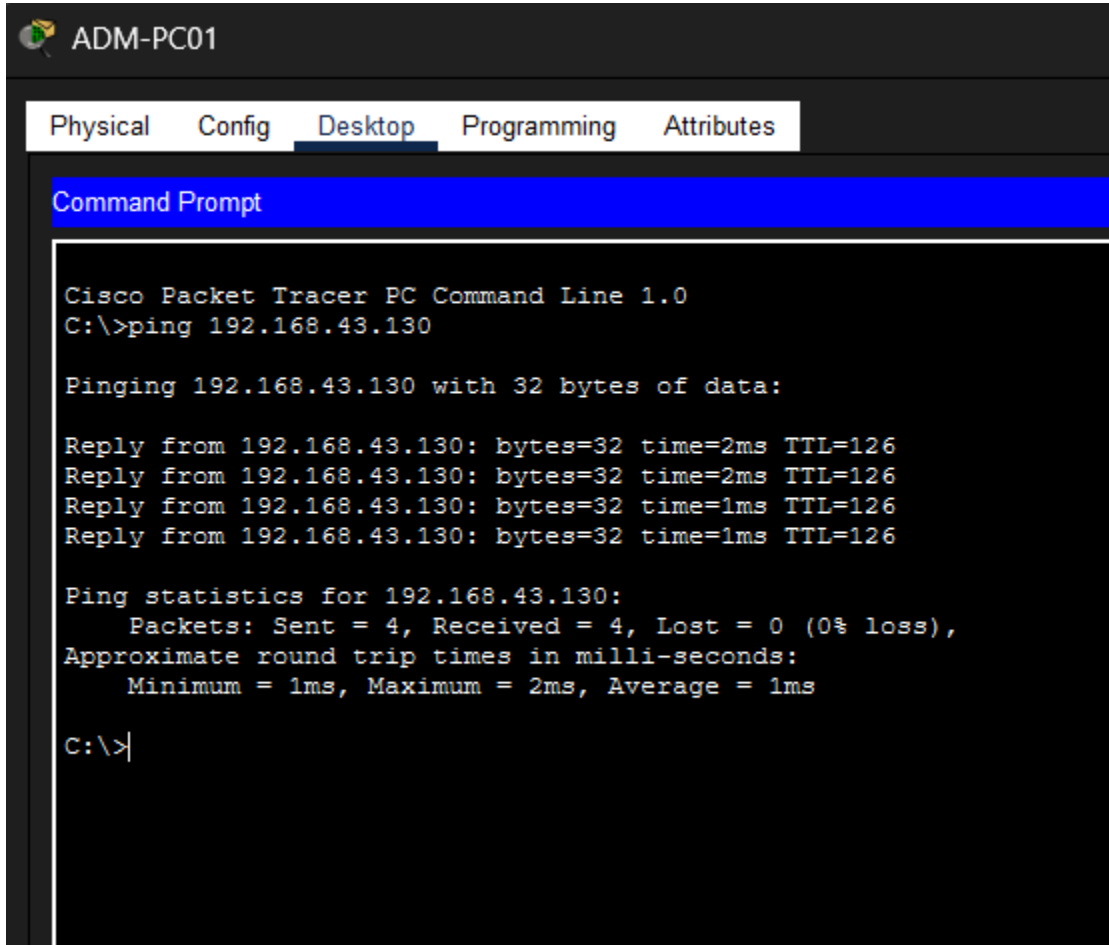
C:\>|

```

Figure 8: LP-PC01 (Licensing & Permits) Connectivity to Shared Printer PRINT-01

## 6.2 Municipal Administration Access

ADM-PC01 was tested against PRINT-01 at 192.168.43.130. The ping returned Sent = 4, Received = 4, Lost = 0 (0% loss), confirming that Municipal Administration also has full IP reachability to the shared printer.

A screenshot of the ADM-PC01 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, showing a 'Command Prompt' window. The command prompt displays the output of a 'ping 192.168.43.130' command. The output shows four successful replies from 192.168.43.130 with varying times (2ms and 1ms) and a TTL of 126. The ping statistics at the bottom confirm that all four packets were sent and received with 0% loss. The command prompt ends with 'C:\>|'.

```
ADM-PC01
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.43.130

Pinging 192.168.43.130 with 32 bytes of data:

Reply from 192.168.43.130: bytes=32 time=2ms TTL=126
Reply from 192.168.43.130: bytes=32 time=2ms TTL=126
Reply from 192.168.43.130: bytes=32 time=1ms TTL=126
Reply from 192.168.43.130: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.43.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>|
```

Figure 9: ADM-PC01 (Municipal Administration) Connectivity to Shared Printer PRINT-01

Both departments selected for CR8 can successfully reach the centralized shared printer. This confirms that PRINT-01 is reachable via the routed static path through MUN-R2 from both authorized VLANs, satisfying the client's change request. These results demonstrate IP-level connectivity to the printer; no physical print job was performed as part of this test.

## 7. Security Constraint Implementation and Verification

The client requires that Public Counter staff (VLAN 10) must not be able to access the Administration network (VLAN 30). This is enforced using a named extended access control list, `PUBLIC_COUNTER_RESTRICTION`, configured on `MUN-R1`.

### 7.1 ACL Logic and Placement

The ACL denies traffic from the Public Counter subnet (192.168.43.0/27) to the Administration subnet (192.168.43.64/27) and permits all other traffic:

```
10 deny ip 192.168.43.0 0.0.0.31 192.168.43.64 0.0.0.31
20 permit ip any any
```

The ACL is applied inbound on interface `GigabitEthernet0/0.10` using `ip access-group PUBLIC_COUNTER_RESTRICTION in`, placing the restriction on traffic as it enters `MUN-R1` from the Public Counter VLAN, before it can be routed to any other network.

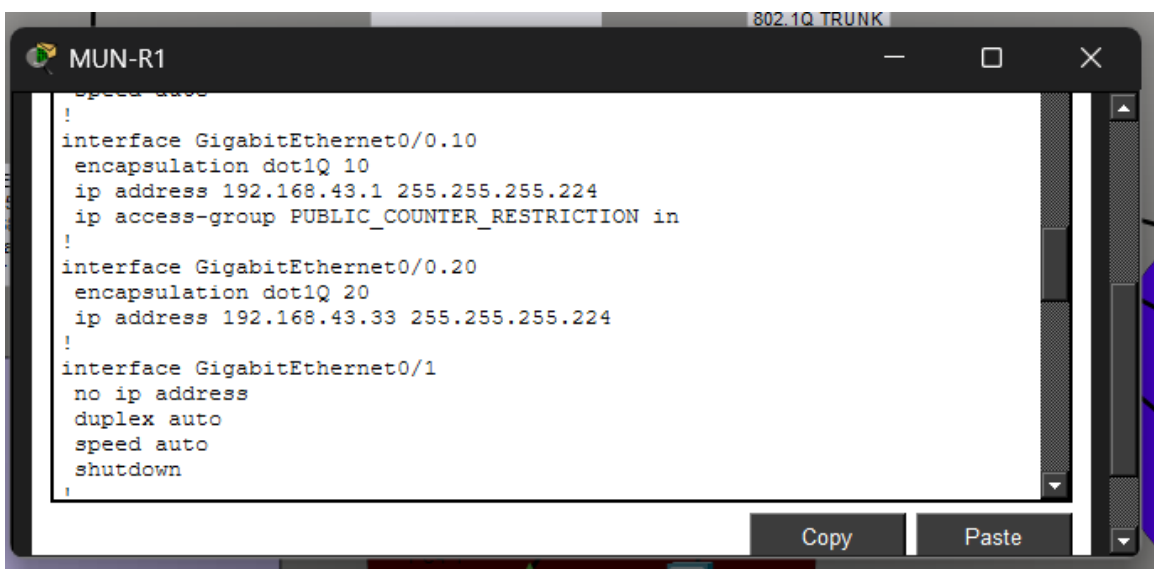


Figure 10: `PUBLIC_COUNTER_RESTRICTION` Applied Inbound on `GigabitEthernet0/0.10`

### 7.2 Blocked Access Test — Public Counter to Administration

`CS-PC01` (VLAN 10, Public Counter) was tested against `ADM-PC01` (192.168.43.66). The result was Reply from 192.168.43.1: Destination host unreachable, with Sent = 4, Received = 0, Lost = 4 (100% loss). This is an expected block rather than a fault: the reply originates from `MUN-R1`'s own gateway interface (192.168.43.1), indicating that the packets reached the router and were then denied by the ACL rather than being lost due to a routing failure.

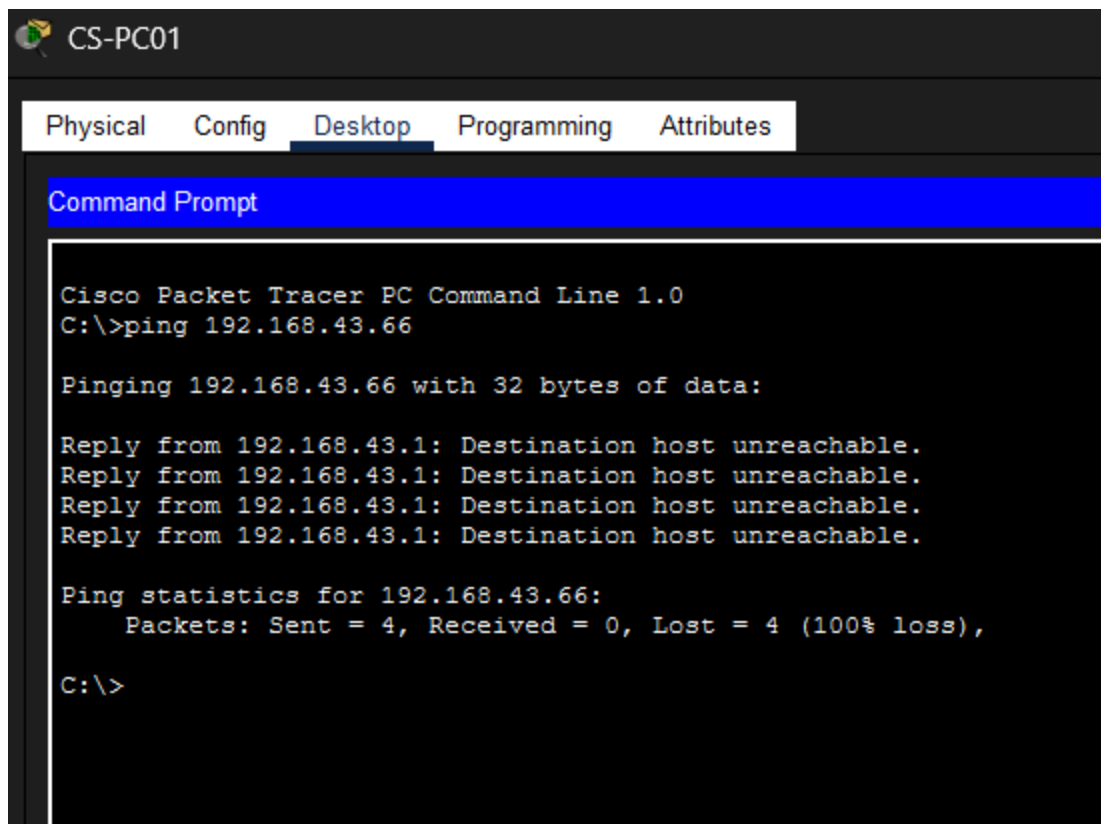


Figure 11: CS-PC01 (Public Counter) Blocked from Reaching Municipal Administration

### 7.3 ACL Match Counter Verification

Following the blocked test, show access-lists on MUN-R1 was used to confirm that the deny statement had actually matched the denied traffic. The output shows 4 match(es) recorded against the deny rule, corresponding to the four packets sent during the CS-PC01 test. This provides router-side confirmation that the block was enforced by the ACL rather than resulting from an unrelated failure.

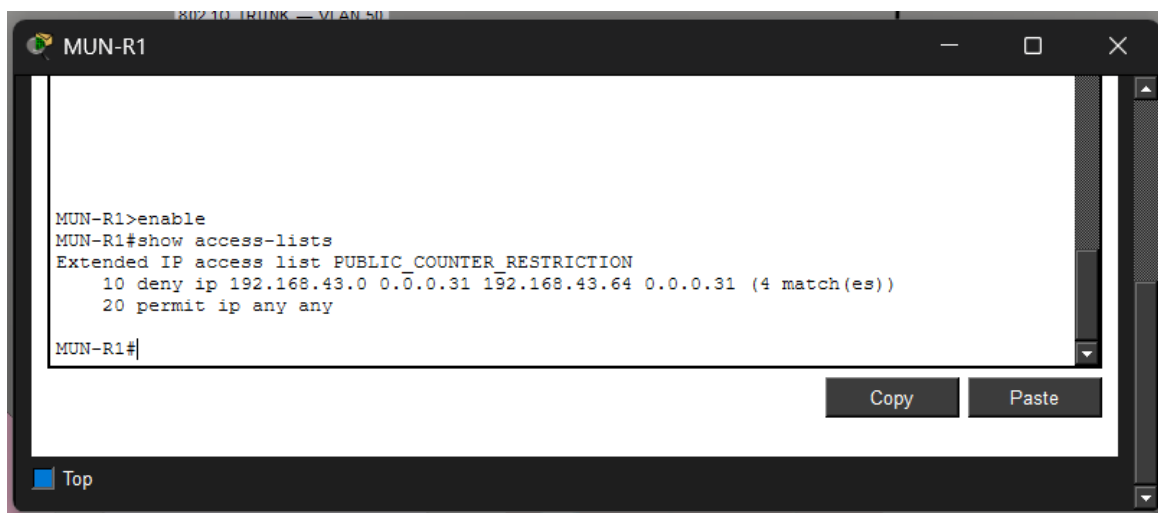
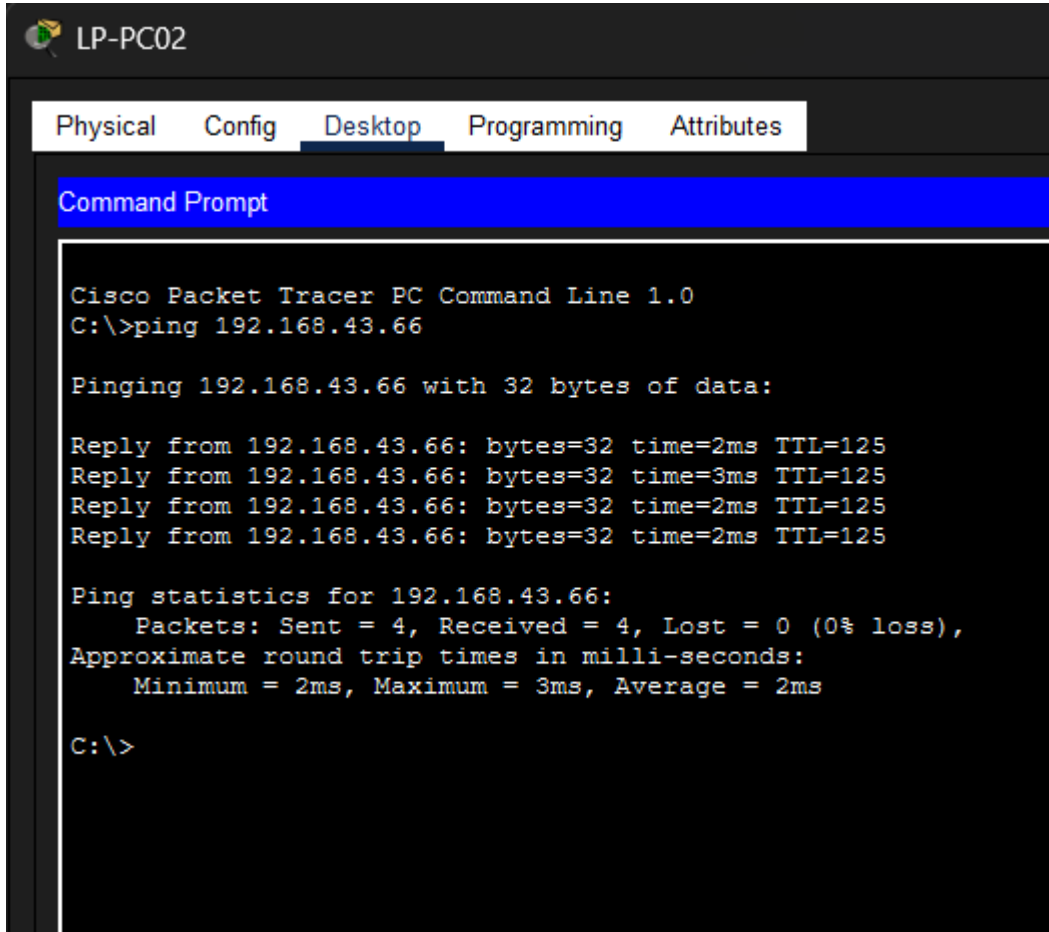


Figure 12: PUBLIC\_COUNTER\_RESTRICTION Deny Rule with Packet Match Counter

## 7.4 Control Test — Authorized Access to Administration

To confirm that Administration was not simply unreachable due to a routing failure, a second test was performed from LP-PC02 (VLAN 20, Licensing & Permits) to ADM-PC01 (192.168.43.66). This traffic is not subject to the deny rule, since the ACL matches only the Public Counter source subnet. The result was Sent = 4, Received = 4, Lost = 0 (0% loss), confirming that Administration is fully reachable from other departments.

The image shows a screenshot of the LP-PC02 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 192.168.43.66. The output indicates that the ping was successful, with 4 packets sent, 4 received, and 0% loss. The round trip times are also displayed: Minimum = 2ms, Maximum = 3ms, and Average = 2ms.

```
LP-PC02
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.43.66

Pinging 192.168.43.66 with 32 bytes of data:

Reply from 192.168.43.66: bytes=32 time=2ms TTL=125
Reply from 192.168.43.66: bytes=32 time=3ms TTL=125
Reply from 192.168.43.66: bytes=32 time=2ms TTL=125
Reply from 192.168.43.66: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.43.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

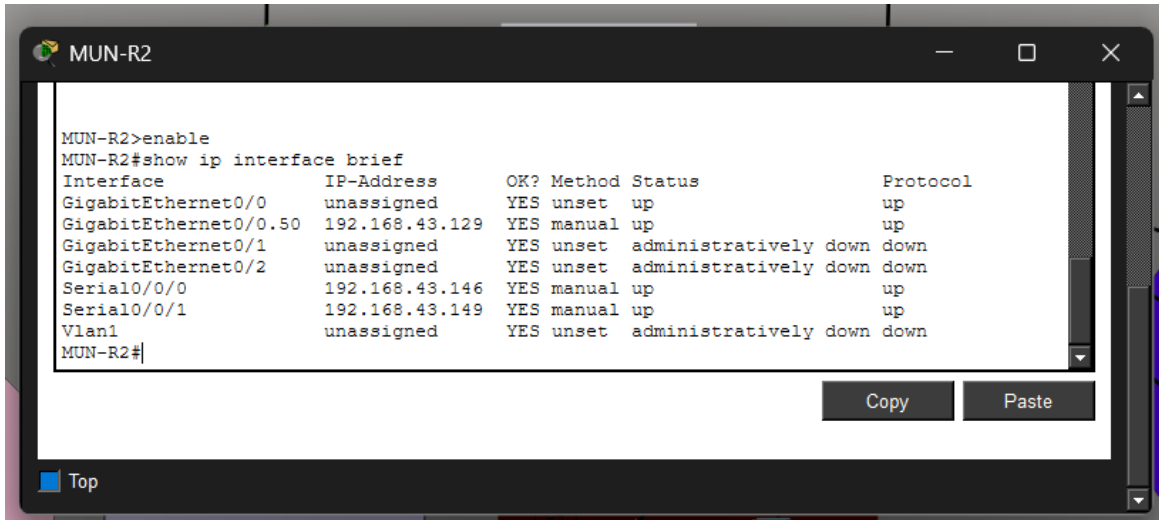
C:\>
```

Figure 13: LP-PC02 (Licensing & Permits) Successfully Reaching Municipal Administration

Together, these results confirm that the security restriction is selective: Public Counter → Administration traffic is blocked, while Licensing & Permits → Administration traffic is permitted, demonstrating that the ACL enforces the client's requirement without over-restricting unrelated traffic.

## 8. Operational Interface Verification

MUN-R2's interfaces were verified using show ip interface brief. GigabitEthernet0/0.50 (192.168.43.129, the VLAN 50 gateway), Serial0/0/0 (192.168.43.146, the link to MUN-R1), and Serial0/0/1 (192.168.43.149, the link to MUN-R3) are all reported as up/up. This confirms that the interfaces required for the Shared Printer Zone and for both inter-router WAN links are operational, supporting the routing and CR8 test results presented above.



```
MUN-R2>enable
MUN-R2#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       unassigned      YES unset    up            up
GigabitEthernet0/0.50    192.168.43.129 YES manual    up            up
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Serial0/0/0              192.168.43.146 YES manual    up            up
Serial0/0/1              192.168.43.149 YES manual    up            up
Vlan1                    unassigned      YES unset    administratively down down
MUN-R2#
```

Figure 14: MUN-R2 Interface Status Confirming VLAN 50 Gateway and WAN Links Up/Up

## 9. Testing Summary

The following table summarizes all tests referenced in this document.

| Test ID | Source   | Destination     | Purpose                        | Expected | Actual   | Status |
|---------|----------|-----------------|--------------------------------|----------|----------|--------|
| T01     | CS-PC01  | VLAN 10 gateway | Local gateway reachability     | Pass     | Pass     | PASS   |
| T02     | LP-PC01  | VLAN 20 gateway | Local gateway reachability     | Pass     | Pass     | PASS   |
| T03     | ADM-PC01 | VLAN 30 gateway | Local gateway reachability     | Pass     | Pass     | PASS   |
| T04     | FIN-PC01 | VLAN 40 gateway | Local gateway reachability     | Pass     | Pass     | PASS   |
| T05     | MUN-R1   | MUN-R2          | WAN connectivity               | Pass     | Pass     | PASS   |
| T06     | MUN-R2   | MUN-R3          | WAN connectivity               | Pass     | Pass     | PASS   |
| T07     | LP-PC01  | PRINT-01        | CR8 printer connectivity       | Pass     | Pass     | PASS   |
| T08     | ADM-PC01 | PRINT-01        | CR8 printer connectivity       | Pass     | Pass     | PASS   |
| T09     | CS-PC01  | ADM-PC01        | Security constraint            | Block    | Blocked  | PASS   |
| T10     | LP-PC02  | ADM-PC01        | Authorized routed connectivity | Pass     | Pass     | PASS   |
| T11     | LP-PC01  | ADM-PC01        | Traceroute — multi-router path | R1→R2→R3 | R1→R2→R3 | PASS   |

## 10. Conclusion

Testing confirms that the network proposed in Milestone 1 has been implemented and operates as intended. VLAN segmentation and 802.1Q trunking are correctly configured across all three switches, and static routing provides the required inter-network reachability across the implemented subnets, subject to the configured security policy. The assigned Multi-Router Path Control challenge is verified by a traceroute showing forwarding across all three routers (MUN-R1 → MUN-R2 → MUN-R3). CR8 — the shared printer zone — has been implemented and confirmed reachable from both authorized departments, Licensing & Permits and Municipal Administration. The client's security constraint is enforced through a named extended ACL on MUN-R1, verified both by a blocked test from the Public Counter VLAN and a selective control test confirming that other departments retain access. The relevant MUN-R2 interfaces supporting these functions are confirmed operational. No functionality beyond that demonstrated by the evidence presented above is claimed.